

Federated Learning-Based Privacy-Preserving Medical Diagnosis System

PHASE – I REPORT

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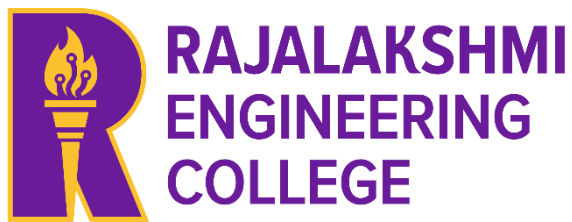
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BONAFIDE CERTIFICATE

Certified that this Report titled “**FEDERATED LEARNING-BASED PRIVACY-PRESERVING MEDICAL DIAGNOSIS SYSTEM**” is the Bonafide work of **SREVARSHA N (230701330)**, **SANJANA R (230701285)**, **NAVEED AHMED BASHA (230701204)**, **PRATHYUSH R (230701240)** who carried out the project work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

“This project addresses the following Sustainable Development Goal(s):

Goal -No: 3 Good Health and Well-Being

Description: Ensure healthy lives and promote well-being for all at all ages.

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ABSTRACT

Diabetes is a chronic disease that requires early detection to support timely intervention and disease management. This project develops a diabetes prediction model using Logistic Regression and compares centralized and federated learning approaches. The study uses clinical attributes including age, BMI, HbA1c level, blood glucose level, hypertension, heart disease, gender, and smoking history. SMOTENC is used to address class imbalance, while federated training uses FedProx-based local optimization with sample-size-weighted server aggregation. Federated experiments are conducted under both IID and non-IID client distributions, with non-IID data generated using a Dirichlet distribution with $\alpha = 0.30$, and the number of clients varied across 3, 5, and 10. A common classification threshold of 0.8752, selected through three-fold out-of-fold validation by maximizing F1-score, is applied across all experiments. Models are evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC, and PR-AUC on the same unseen test set. The centralized model achieved 95.51% accuracy, while the three-client non-IID federated model achieved 95.65% accuracy, showing comparable predictive performance. The results further show that IID performance remained relatively stable across client counts, whereas non-IID configurations exhibited greater variation in predictive performance.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
AUC	Area Under the Curve
BMI	Body Mass Index
CSV	Comma-Separated Values
FedAvg	Federated Averaging
FedProx	Federated Proximal
FL	Federated Learning
FN	False Negative
FP	False Positive
HbA1c	Glycated Haemoglobin
IID	Independent and Identically Distributed
IoT	Internet of Things
L2	L2 Regularization
LR	Logistic Regression
ML	Machine Learning
MRI	Magnetic Resonance Imaging
OOF	Out-of-Fold

PR-AUC	Precision-Recall Area Under the Curve
ROC-AUC	Receiver Operating Characteristic – Area Under the Curve
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
SMOTENC	Synthetic Minority Over-sampling Technique for Nominal and Continuous
TN	True Negative
TP	True Positive
UML	Unified Modeling Language

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Diabetes is a chronic condition with high blood glucose levels. Early diabetes detection enables proper treatment and reduces the risk of severe complications. With the growing availability of healthcare data, machine learning can extract useful insights from patient data to predict diabetes. These algorithms identify patterns among individuals with and without diabetes and build models to classify individuals based on their characteristics.

This project uses Logistic Regression for diabetes prediction. The algorithm uses attributes such as age, BMI, HbA1c level, blood glucose level, hypertension, heart disease, gender, and smoking history. Data preprocessing and class-balancing techniques are applied to prepare the data for model training.

The diabetes prediction task is also explored using federated learning. In the federated setting, clients train the model using their local data without transferring their raw training data to the server. FedProx is used for local optimization, while the server aggregates the client model parameters using sample-size-weighted averaging. Several clients hold different subsets of the training data, and the experiments consider both IID and non-IID data distributions. The project also studies how the number of clients affects model performance.

1.2 OBJECTIVE

The proposed project aims to develop and evaluate a diabetes prediction system using Logistic Regression in both centralized and federated learning environments. The project focuses on comparing centralized and federated prediction while investigating the effects of client-level data distribution.

The main objectives of the project are:

1. To develop a centralized diabetes prediction model using Logistic Regression that can predict whether a person is likely to have diabetes based on the medical attributes provided. The system also uses SHAP-based explanations to help interpret the contribution of the input features to the prediction.
2. To develop a federated diabetes prediction model using Logistic Regression with FedProx-based local optimization. The primary federated comparison is between the centralized model and the three-client non-IID federated model, using the same evaluation dataset and the same fixed classification threshold. This allows the predictive performance of centralized and federated learning to be compared under a heterogeneous client setting.
3. To investigate client-level statistical heterogeneity by increasing the number of federated clients from 3 to 5 and 10 under both IID and non-IID settings. The objective is to study how client-level data distributions affect model performance, examine the effect of federation size, and compare the behavior of IID and non-IID federated learning.
4. To evaluate classification threshold selection by comparing the conventional 0.50 threshold with the custom fixed threshold selected through three-fold out-of-fold validation. The objective is to examine the resulting precision-recall trade-off and justify the use of a common threshold for comparing the centralized and federated models.

1.3 EXISTING SYSTEM

Traditional diabetes prediction relies on centralized machine learning, pooling all data into a central database for training. This approach gives the learning algorithm full access to the training set, learning from all records

simultaneously. Although simple and accurate, it necessitates aggregating data for training.

In healthcare, centralizing data can be problematic, as patient information from various sources must be combined for a single model. Patient data confidentiality makes pooling data centrally for training challenging. Hence, distributed methods are appealing, as they avoid sharing client data among participants.

Another factor affecting centralized predictive performance is class imbalance in training data. Non-diabetic records often outnumber diabetic ones, leading to a model biased toward the majority class. This causes poor minority-class recall, undesirable for this application. To optimize the model, preprocessing and appropriate class-balancing techniques should be applied.

The current method does not explicitly address client data heterogeneity or participant count. This research proposes a federated learning method to examine the effects of IID and non-IID data.

1.4 PROPOSED SYSTEM

The proposed system aims to develop a diabetes prediction framework using Logistic Regression with both centralized and federated learning approaches. The dataset is first cleaned by removing duplicate and invalid records, standardizing numerical attributes, and encoding categorical variables. The cleaned data is then divided into training and testing sets using an 80:20 stratified split, maintaining a similar target-class distribution in both sets.

SMOTENC is applied only to the training data to address class imbalance between diabetic and non-diabetic records. Numerical features are standardized, while categorical and binary features are encoded in a form suitable for SMOTENC and Logistic Regression. The test set remains untouched throughout preprocessing and training to avoid data leakage.

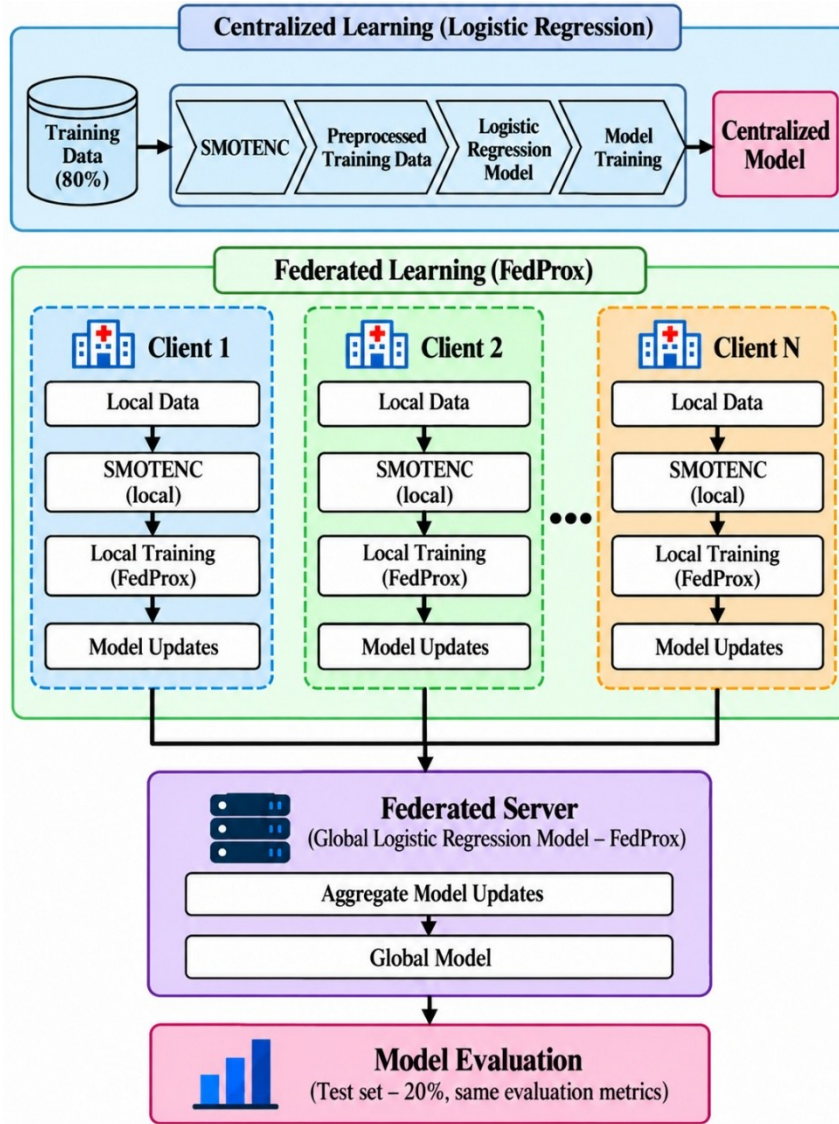


Fig 1.1. Overview of the proposed diabetes prediction system

The study uses both centralized and federated learning. The centralized approach trains a Logistic Regression model using the complete training data. In the federated approach, the training data is distributed among multiple clients. Each client performs local preprocessing, applies SMOTENC, and trains the model using FedProx-based local optimization. The server then combines the client model parameters using sample-size-weighted averaging, without receiving the clients' raw training data.

Federated experiments are conducted under both IID and non-IID settings. IID data is distributed using stratified allocation, while non-IID data is generated

using a Dirichlet distribution with $\alpha = 0.30$. The number of clients is varied across 3, 5, and 10 to study the effect of client-level data distributions on model performance.

A three-fold out-of-fold validation strategy is used to select a common classification threshold by maximizing the F1-score. The resulting threshold of 0.8752 is fixed and applied across the centralized and federated experiments. The conventional 0.50 threshold is also evaluated as a reference. Model performance is compared using Accuracy, Precision, Recall, F1-score, ROC-AUC, PR-AUC, and confusion matrices.

CHAPTER 2

LITERATURE SURVEY

This section of the paper, provides an all-round review about multiple existing systems that holds similar objective and problem statements as that of the proposed project. The study of these existing papers will produce valuable insights for the construction of the proposed system.

2.1 LITERATURE REVIEW

Dash et al. [1] study privacy-preserving diabetes and heart disease prediction using federated learning. Their Federated Learning with Weighted Conglomeration Optimization (FLWCO) method selects local classifiers based on their performance and uses confidence-score thresholds to improve communication efficiency and convergence. However, the study mainly focuses on aggregation and optimization rather than examining how different client data distributions affect the federated model.

Grant et al. [2] compare centralized and federated learning for diabetes prediction using the PIMA Indians Diabetes Dataset and Logistic Regression. Their federated model achieves an ROC-AUC of 0.77, compared with 0.78 for the centralized model. This provides a useful comparison between centralized and federated learning on structured diabetes data. However, the study does not investigate non-IID client distributions or different numbers of participating clients.

Haripriya et al. [3] propose a federated learning approach with adaptive aggregation for medical image classification. Their method dynamically switches between FedAvg and FedSGD based on client weight variations and evaluates medical image datasets including chest X-rays, brain MRIs, and diabetic retinopathy data. The study uses a ten-client setup and also considers

larger client configurations, highlighting communication and scalability issues. However, it does not provide the same controlled comparison of IID and non-IID client distributions used in the proposed work.

Demir et al. [4] investigate privacy-preserving diabetes and heart disease prediction using federated learning. Their Data Unaware Classification Based on Association (duCBA) approach aggregates logical relational rules instead of high-dimensional model weights, improving computational efficiency and interpretability. However, statistical heterogeneity between clients is not addressed.

Atmoko et al. [5] introduce FedXChain, an explainable federated learning framework for heterogeneous and non-IID healthcare data. The framework combines SHAP-based aggregation, adaptive trust-weighted aggregation, and blockchain-based auditing. Using the Wisconsin Breast Cancer dataset, their Logistic Regression model achieves 96.5% accuracy. However, the evaluation uses a fixed ten-client configuration and does not compare the federated model with an equivalent centralized model.

Bebortta et al. [6] propose FedEHR for privacy-preserving cardiac risk prediction using IoT-enabled electronic health records. Their framework combines a sparse Support Vector Machine, cluster Primal-Dual Splitting, and a round-robin heuristic to improve prediction while reducing communication and computational requirements. Although the study identifies data quality, heterogeneity, and scalability as important issues, it does not experimentally examine how client-level data distributions affect the difference between centralized and federated performance.

Shahid et al. [7] propose Health-FedNet for privacy-preserving chronic disease prediction. Their framework combines differential privacy, Paillier homomorphic encryption, and adaptive node weighting. The study also investigates different numbers of clients, but its focus is mainly on privacy and

secure federated learning rather than a controlled comparison of IID and non-IID distributions against a centralized baseline.

Rahman et al. [8] develop a federated IoT-based framework for early stroke-risk prediction and continuous patient monitoring. Their approach uses FedAvg together with wearable sensing and multiple feature-selection and optimization techniques. The study achieves high prediction accuracy, but its evaluation uses stratified IID partitions, leaving non-IID client variation for future investigation.

Sorayaie Azar et al. [9] investigate privacy-preserving coronary artery disease prediction using the FMLCA framework. Their approach combines federated machine learning with k-anonymity and SHAP-based explainability. The study achieves 83.21% classification accuracy, but evaluation is performed on a relatively small clinical dataset, leaving performance under diverse and heterogeneous multi-centre settings unexplored.

Amin et al. [10] propose Distribution-Aware Federated Learning (DA-FL) for diabetes prediction using the CDC BRFSS 2021 dataset. Their method addresses non-IID data and class imbalance through class-weighted local training and distribution-aware aggregation. However, the study uses a fixed five-client federation and does not extensively investigate how changing the number of clients affects performance under IID and non-IID conditions.

2.2 COMPARISON AND DISCUSSION

The reviewed studies show that federated learning in healthcare has mainly focused on improving aggregation, optimization, privacy, security, communication efficiency, and explainability. Many studies use a fixed number of clients, while relatively few examine how differences in client-level data distributions affect the performance gap between centralized and federated learning.

Some studies directly compare centralized and federated learning, such as Grant et al. [2], while others investigate federated methods mainly against other federated baselines. Studies such as Atmoko et al. [5] and Amin et al. [10] consider heterogeneous or non-IID data, but their experiments do not provide the same combination of centralized comparison, IID versus non-IID analysis, and varying client counts used in this project.

Class imbalance is another important consideration in diabetes prediction. Existing studies use different approaches such as SMOTE, class weighting, and distribution-aware training. The proposed system uses SMOTENC to handle class imbalance while retaining the same Logistic Regression prediction model across centralized and federated experiments.

Based on the reviewed literature, there is limited work that simultaneously compares centralized and federated learning, evaluates IID and non-IID client distributions, varies the number of clients across 3, 5, and 10, and considers class imbalance under a common evaluation framework. The proposed project addresses this gap by maintaining the same Logistic Regression-based prediction methodology while systematically varying client distribution and federation size.

2.3 CONCLUSION

The reviewed studies demonstrate the potential of federated learning for healthcare prediction while highlighting the importance of privacy, optimization, aggregation, and heterogeneous client data. However, relatively few studies provide a controlled comparison between centralized and federated learning while simultaneously examining IID and non-IID client distributions and different numbers of clients.

Therefore, this project uses Logistic Regression with SMOTENC for diabetes prediction and extends it to federated learning using FedProx. The study compares centralized learning with IID and non-IID federated settings using a

common evaluation procedure and examines client counts of 3, 5, and 10. This provides a controlled way to study how client-level statistical heterogeneity and federation size are associated with federated diabetes-prediction performance.

CHAPTER 3

SYSTEM DESIGN

This chapter of the project work briefly describes the design of the proposed diabetes prediction system using the system architecture and the model diagrams. Additionally, the system's requirements, including software, hardware, and dataset requirements, are also described. This chapter also elaborates on the overall architecture of the proposed system and the data flow through the centralized and federated learning environments.

3.1 SYSTEM ARCHITECTURE

The proposed system performs diabetes prediction using Logistic Regression while examining the effect of distributed learning on model performance. The architecture begins with the Diabetes dataset, which is cleaned and validated before being divided into training and testing sets using a stratified 80:20 split. The training data undergoes numerical scaling, categorical encoding, and SMOTENC processing to address class imbalance. The test set remains untouched during preprocessing, training, and threshold selection.

The proposed architecture consists of two major learning approaches: centralized learning and federated learning. In centralized learning, the complete training dataset is processed as a single dataset and used to train a Logistic Regression model. In federated learning, the training data is distributed among multiple clients. Each client preprocesses its local data, performs class balancing using SMOTENC, and trains the Logistic Regression model using FedProx-based local optimization. The resulting model parameters are sent to the server, where they are combined using sample-size-weighted averaging.

The architecture of the proposed diabetes prediction system is shown in Fig 3.1 below, illustrating the flow from dataset preparation to final model evaluation.

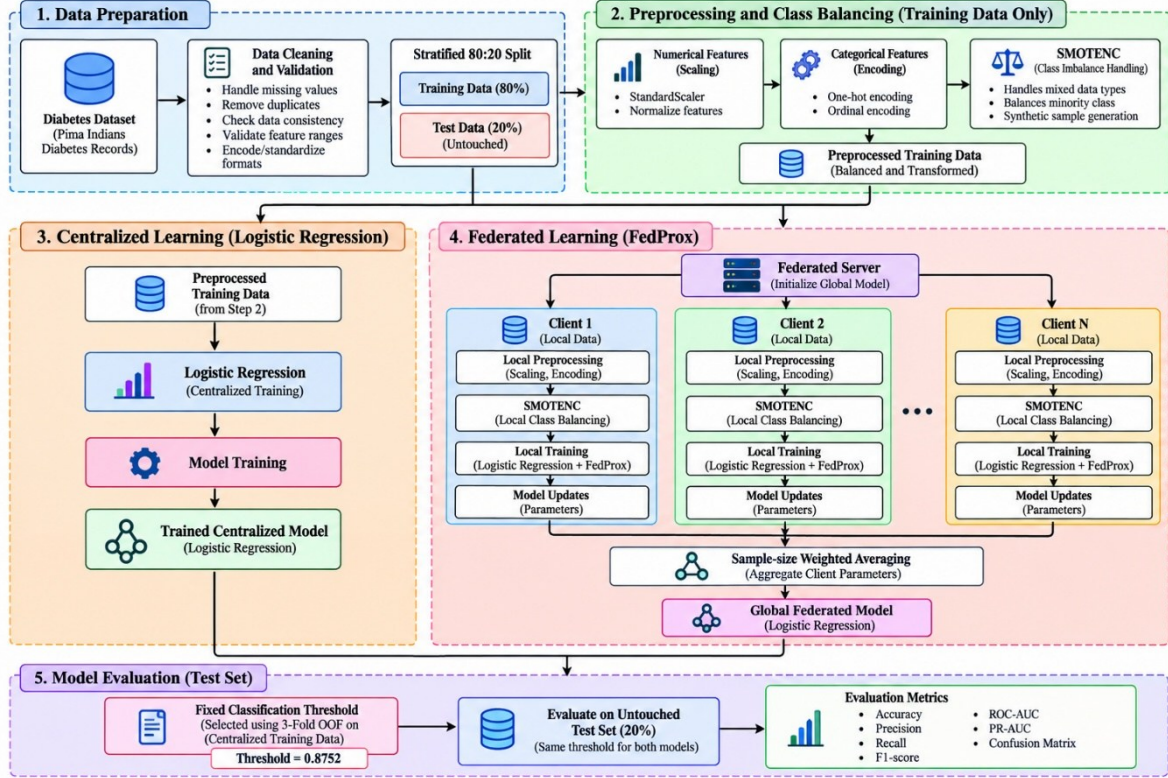


Fig 3.1. Architecture diagram

As shown in the diagram, the process begins with the Diabetes dataset, which is cleaned and validated before being divided into training and testing data. The training data then undergoes feature preprocessing and class balancing using SMOTENC. The processed training data is subsequently directed into either the centralized or federated learning pipeline.

In the centralized approach, the processed training data is used to train the Logistic Regression model. For the federated approach, the training data is distributed among multiple clients according to IID or non-IID configurations. Each client performs local training using FedProx, after which the server aggregates the client parameters using sample-size-weighted averaging. Federated experiments are performed with 3, 5, and 10 clients to investigate the effect of client-level data distributions and federation size.

Finally, a three-fold out-of-fold validation procedure is used on the centralized training data to select the classification threshold that maximizes F1-

score. The resulting threshold of 0.8752 is fixed across the centralized and federated experiments. The final models are evaluated on the untouched test set using Accuracy, Precision, Recall, F1-score, ROC-AUC, PR-AUC, and confusion matrix.

3.2 SYSTEM REQUIREMENTS

This section describes the software, hardware, dataset, and deployment requirements for the proposed diabetes prediction system. The requirements support the implementation of centralized Logistic Regression, SMOTENC-based class balancing, and federated learning using FedProx.

3.2.1 SOFTWARE REQUIREMENTS

Python is used as the primary programming language because it provides the required libraries for data preprocessing, machine learning, numerical computation, and evaluation. The centralized model uses Logistic Regression, while the federated implementation uses a custom Logistic Regression optimization procedure to incorporate the FedProx proximal term into local training.

Software Component	Suggested Version	Purpose
Python	3.10 or higher	Primary programming language
Pandas	Compatible version	Dataset loading and manipulation
NumPy	Compatible version	Numerical computation
Scikit-learn	Compatible version	Preprocessing, Logistic Regression and evaluation
Imbalanced-learn	Compatible version	SMOTENC-based class balancing
Matplotlib	Compatible version	Visualization of experimental results
Jupyter Notebook / VS Code	Compatible version	Development and experimentation
Streamlit	Compatible version	Presentation of final results

Table 3.1. Software Requirements

The system can be built in a Python coding environment using tools like Anaconda or Visual Studio Code. The required libraries will provide the needed features for data preprocessing, balancing training data, implementing classification models, and analyzing results.

3.2.2 HARDWARE REQUIREMENTS

The proposed system does not require specialized high-performance hardware because Logistic Regression is computationally efficient for structured tabular data. However, sufficient memory and processing capability are required for SMOTENC resampling and repeated federated experiments involving multiple clients and communication rounds.

Hardware Component	Suggested Specification	Purpose
Processor	Intel Core i5 / AMD Ryzen 5 or higher	Model training and data processing
RAM	8 GB or higher	Handling dataset and intermediate data
Storage	10 GB or higher available space	Dataset, source code and results
GPU	Optional	Not required for Logistic Regression
Operating System	Windows 10/11 or Linux	System development and execution
Network	Stable internet connection	Package installation and deployment

Table 3.2. Hardware Requirements

The experiments can be executed on a standard development machine, which lacks a graphics processing unit (GPU). Additional computational resources may be utilized depending on the number of clients or repetitions of federated experiments.

3.2.3 DATASET REQUIREMENTS

The system uses the Diabetes Prediction Dataset for all centralized and federated experiments. The target variable is diabetes, while the input attributes include age, gender, BMI, hypertension, heart disease, smoking history, HbA1c level, and blood glucose level.

The original dataset contains 100,000 records. After removing duplicate records, standardizing categorical text values, merging the "ever" smoking-history category into "former", and performing basic validity checks, the cleaned dataset contains 96,146 records.

The cleaned dataset is divided into training and testing portions using a stratified 80:20 split with `random_state = 42`. All preprocessing and class-balancing operations are performed only on the training data. The test data remains untouched until final evaluation.

Numerical features are standardized using `StandardScaler`, while binary and categorical features are encoded for `SMOTENC` processing. `SMOTENC` is applied only to the training data to address class imbalance. In the federated setting, it is performed independently at each client, and the resulting synthetic samples remain local to that client.

3.2.4 DEPLOYMENT AND SCALING

The proposed system is primarily designed as an experimental platform for comparing centralized and federated diabetes prediction. In the centralized setting, the complete training dataset is processed within a single environment, whereas in the federated setting, the training data is distributed among multiple clients.

The federated system is evaluated using 3, 5, and 10 clients under both IID and non-IID conditions. IID data is generated through stratified allocation,

while non-IID data is generated using class-wise Dirichlet partitioning with $\alpha = 0.30$. The remaining training and evaluation settings are kept consistent across the experiments.

3.3 SYSTEM DESIGN

This section describes the system design using UML-based diagrams to illustrate the operation of the proposed diabetes prediction system. The diagrams represent the major processes involved in data preparation, model training, prediction, and evaluation.

3.3.1 ACTIVITY DIAGRAM

An activity diagram represents the sequence of activities performed by the proposed diabetes prediction system. The overall workflow of the system is illustrated in Fig 3.2.

The process begins by loading the Diabetes dataset, followed by data cleaning and validation. The cleaned dataset is then divided into training and testing subsets using a stratified 80:20 split. The training data undergoes feature preprocessing and class balancing using SMOTENC, while the test data remains untouched.

The learning phase then provides two alternatives: centralized learning or federated learning. In the centralized approach, the complete training data is used to train the Logistic Regression model. In the federated approach, the training data is distributed among clients using either IID or non-IID partitioning. Each client performs local preprocessing and SMOTENC and trains the Logistic Regression model using FedProx-based local optimization.

The trained client parameters are sent to the server and combined using sample-size-weighted averaging to obtain the global model. Three-fold out-of-

fold predictions from the centralized training data are used to select the common classification threshold by maximizing F1-score. The selected threshold of 0.8752 is then applied to the untouched test data, and the final performance is evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC, PR-AUC, and the confusion matrix.

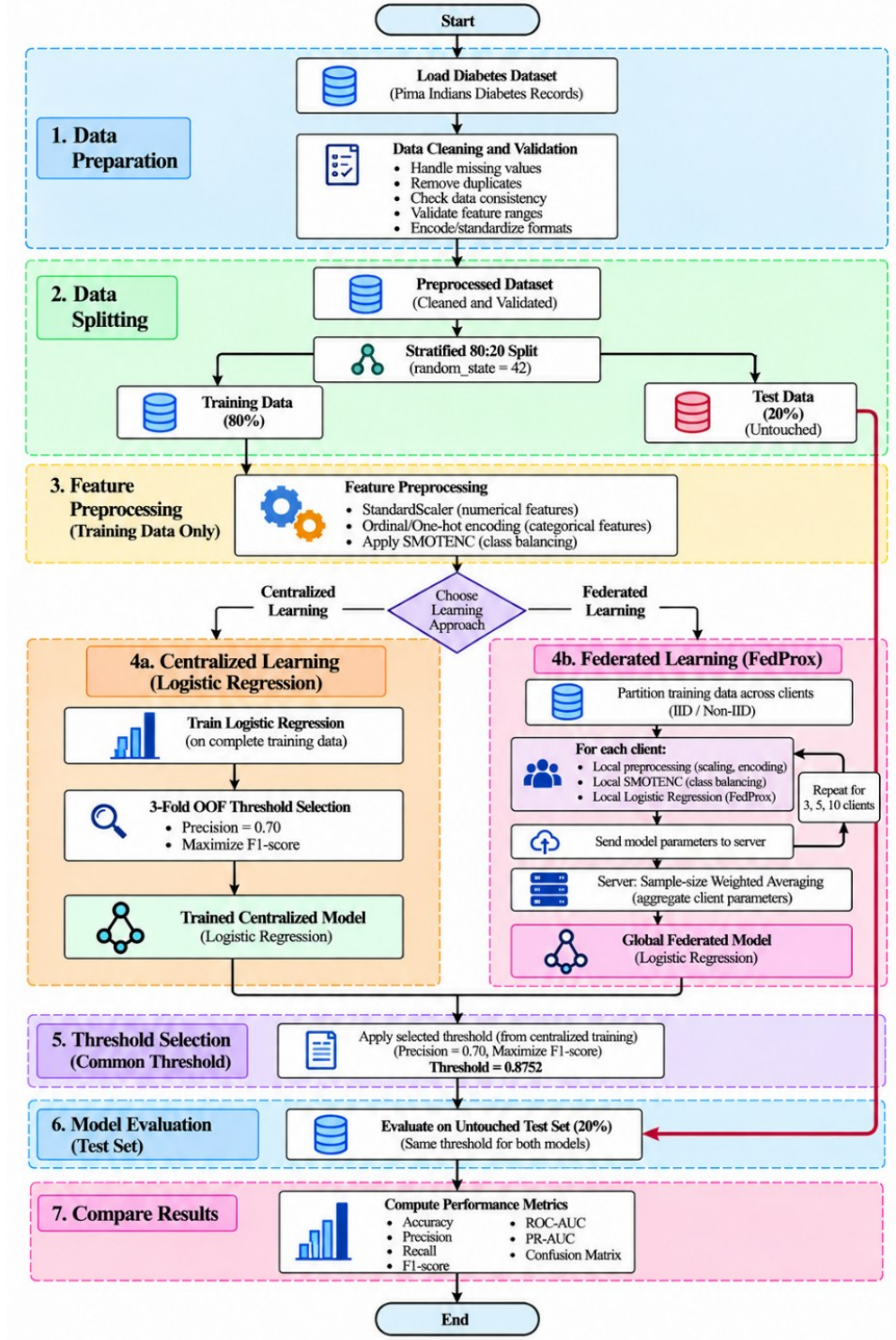


Fig 3.2. Activity Diagram

3.3.2 DATA FLOW DIAGRAM

A data flow diagram represents how data moves between the different components of the proposed system throughout its lifecycle. The complete data flow of the proposed system is illustrated in Fig 3.3.

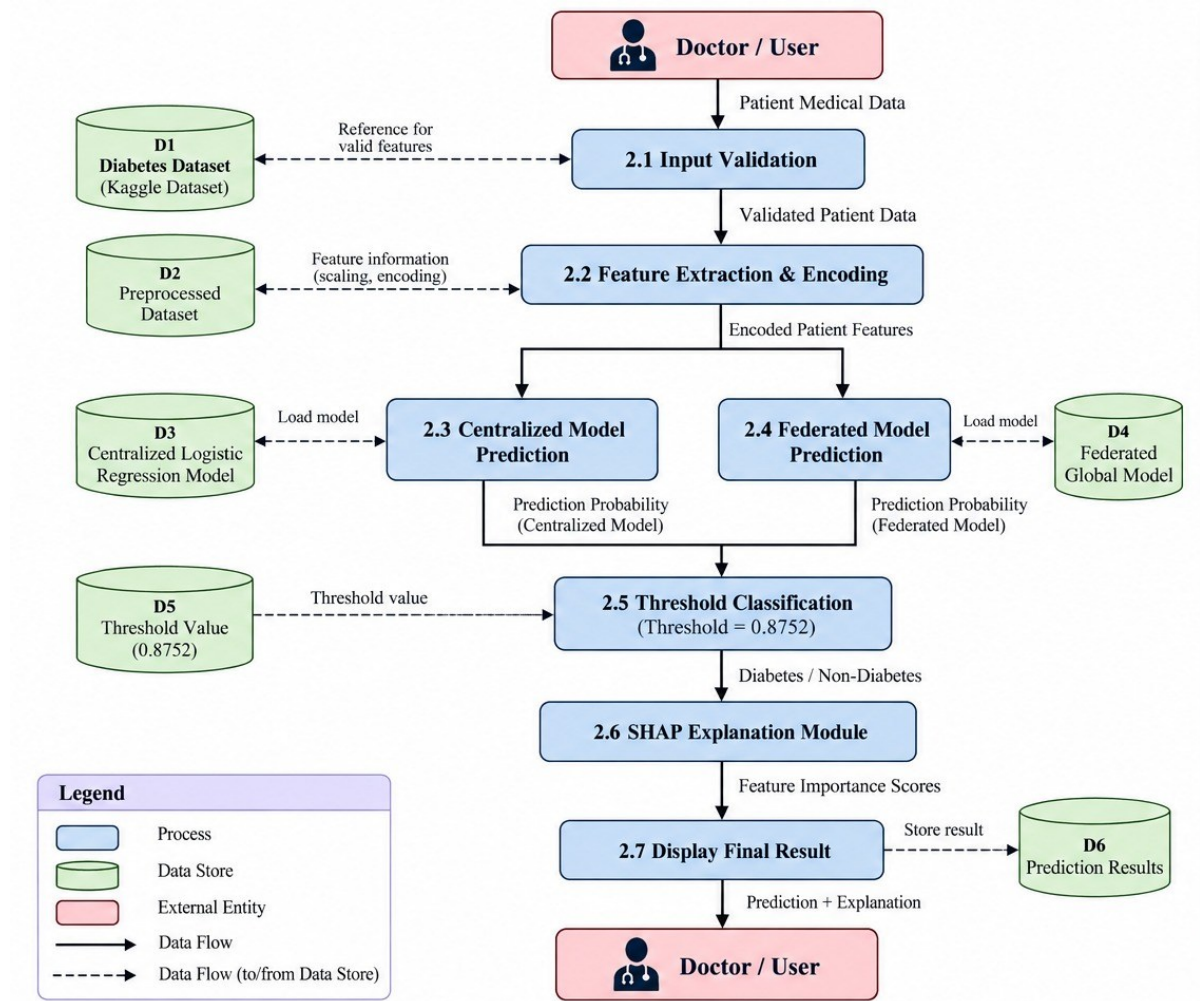


Fig 3.3. Data Flow diagram

The data flow begins with the Diabetes Prediction Dataset, which undergoes cleaning and validation. The cleaned data is divided into training and testing subsets using a stratified 80:20 split. The training data then undergoes feature preprocessing and SMOTENC-based class balancing, while the test data is kept separate for final evaluation.

In the centralized learning pipeline, the processed training data is directly used to train the Logistic Regression model.

In the federated learning pipeline, the training data is distributed among multiple clients according to the selected IID or non-IID configuration. Each client performs local preprocessing, applies SMOTENC, and trains the Logistic Regression model using FedProx. The resulting model parameters are transmitted to the federated server, where they are combined using sample-size-weighted averaging to produce the global model.

The trained centralized and federated models generate prediction probabilities for the test samples. The common threshold of 0.8752, selected using three-fold out-of-fold validation on the centralized training data, is applied to obtain the final class predictions. The resulting predictions are evaluated using the selected performance metrics.

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGIES

The project develops and evaluates a diabetes prediction system using Logistic Regression in both centralized and federated learning environments. The methodology includes data preparation, centralized model development, federated model training, client-level data distribution, diabetes prediction, and model evaluation.

The study first prepares the Diabetes Prediction Dataset and addresses class imbalance using SMOTENC. A centralized Logistic Regression model is developed as the baseline. A corresponding federated Logistic Regression model is developed using FedProx-based local optimization and sample-size-weighted server aggregation.

The federated experiments consider both IID and non-IID client distributions. The non-IID distributions are generated using class-wise Dirichlet partitioning with $\alpha = 0.30$. The number of clients is varied across 3, 5, and 10 to investigate the behaviour of the federated model under different federation sizes.

A common classification threshold is selected using three-fold out-of-fold validation on the centralized training data and is then fixed across the experiments. The centralized and federated models are evaluated on the same unseen test dataset using Accuracy, Precision, Recall, F1-score, ROC-AUC, PR-AUC, and confusion matrices.

4.2 MODULES

The proposed system is divided into the following modules:

1. Data Preparation Module

2. Centralized Training Module
3. Federated Training Module
4. Client Management Module
5. Diabetes Prediction Module
6. Results and Visualization Module

4.2.1 DATA PREPARATION MODULE

The Data Preparation Module is responsible for preparing the Diabetes Prediction Dataset for model development. It handles data cleaning, validation, feature preprocessing, train-test splitting, and class imbalance handling.

The module prepares the numerical, binary, and categorical attributes in a form suitable for the Logistic Regression models. It also performs class balancing using SMOTENC on the training data while keeping the test data separate for final evaluation.

4.2.2 CENTRALIZED LEARNING MODULE

The Centralized Training Module develops the baseline diabetes prediction model using Logistic Regression. It uses the prepared training data to train a single model with all available centralized training records.

This module also determines the common classification threshold used for the experimental comparison. The selected threshold is subsequently fixed so that the centralized and federated models are evaluated using the same decision rule.

4.2.3 FEDERATED LEARNING MODULE

The Federated Training Module develops the diabetes prediction model in a distributed learning environment. Instead of combining the training records at a

central location, individual clients perform model training using their respective local datasets.

The module uses FedProx-based local optimization and combines the locally trained model parameters at the federated server using sample-size-weighted averaging. Both IID and non-IID federated training configurations are supported.

4.2.4 CLIENT MANAGEMENT MODULE

The Client Management Module is responsible for creating the federated clients and controlling the distribution of training data among them. The module is used to investigate how different client-level data distributions and federation sizes influence the performance of the federated diabetes prediction model.

Two types of client distributions are implemented: IID and non-IID.

Non-IID Client Distribution

The Client Management Module is responsible for creating and managing the federated clients and distributing the training data among them.

The module supports both IID and non-IID client distributions. It also allows the number of participating clients to be varied between 3, 5, and 10, enabling the effect of client-level data distributions and federation size to be investigated.

4.2.5 DIABETES PREDICTION MODULE

The Diabetes Prediction Module provides the user-facing diabetes prediction functionality through a Streamlit application. The application allows a user to upload a medical report, extracts the required patient attributes, and provides a diabetes prediction using the trained model.

The application supports prediction using both the centralized and federated models. It also provides a SHAP-based explanation to help interpret the contribution of the input features to the prediction.

4.2.6 RESULTS AND VISUALIZATION MODULE

The Results and Visualization Module evaluates the trained centralized and federated models and presents the experimental results in a form that allows direct comparison.

The evaluation incorporates Accuracy, Precision, Recall, F1-score, ROC-AUC, PR-AUC, and the confusion matrix. The principal classification metrics are derived from the equations presented below.

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)}$$

$$Precision = \frac{TP}{(TP + FP)}$$

$$Recall = \frac{TP}{(TP + FN)}$$

$$F1 - score = 2 \times \frac{(Precision \times Recall)}{(Precision + Recall)}$$

In this context, TP denotes true positives, TN denotes true negatives, FP denotes false positives, and FN denotes false negatives.

The centralized model provides the reference baseline, while the three-client IID and non-IID federated models are compared against this baseline under the same evaluation procedure. The study also compares federated performance across 3, 5, and 10 clients under both IID and non-IID distributions.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION RESULTS

The implementation of the proposed diabetes prediction system and the corresponding experimental analysis are presented in this chapter. The implementation covers data preparation, centralized Logistic Regression, federated learning, client-level data distribution, diabetes prediction, and model evaluation. The experiments are designed to compare centralized and federated learning under consistent preprocessing and evaluation conditions, while also examining IID and non-IID client distributions and different numbers of federated clients.

5.1.1 IMPLEMENTATION OF DATA PRE-PROCESSING

The implementation begins with the Diabetes Prediction Dataset from Kaggle, which originally contains 100,000 records and 9 attributes consisting of numerical, binary, and categorical features. Duplicate records are removed, categorical text values are standardized, and the ever category in `smoking_history` is merged with former. Sanity checks are performed on `age`, `bmi`, `HbA1c_level`, and `blood_glucose_level`, along with checks for missing values. After cleaning, the dataset contains 96,146 records.

The cleaned dataset is divided into training and testing subsets using an 80:20 stratified split with `random_state = 42`. This produces 76,916 training records and 19,230 test records. Stratification maintains the target-class distribution across the two subsets. All subsequent preprocessing and class-balancing operations are performed only on the training data, while the test set remains untouched until final evaluation.

The numerical features — age, bmi, HbA1c_level, and blood_glucose_level — are standardized using StandardScaler. The binary features hypertension and heart_disease, together with gender and smoking_history, are ordinal encoded for the SMOTENC processing stage. This produces an intermediate eight-feature representation with four continuous and four categorical features.

To address class imbalance, SMOTENC is applied to the training data using categorical feature positions [4, 5, 6, 7]. In the centralized setting, SMOTENC is applied to the complete training data and separately within the folds used for OOF threshold selection. In the federated setting, SMOTENC is applied independently by each client, with the resulting synthetic samples remaining local to that client. Clients with fewer than six minority samples skip local oversampling.

After SMOTENC, binary and categorical features are converted using one-hot encoding with drop="first", while the standardized numerical features are retained. The resulting representation is supplied to the Logistic Regression model. The test set does not undergo SMOTENC or any other resampling operation.

5.1.2 IMPLEMENTATION OF CENTRALIZED LOGISTIC REGRESSION

The centralized implementation is developed as the baseline model for the study. Logistic Regression is trained using the preprocessed and SMOTENC-balanced training dataset. The implementation uses `max_iter = 1000` and `random_state = 42`.

Fixed Threshold Selection

To determine a common classification threshold without using the test set, a three-fold stratified Out-of-Fold (OOF) procedure is applied to the centralized

training data.

For every fold, preprocessing, SMOTENC, model training, and probability prediction are performed using the training portion of the fold. Predictions are then generated for the corresponding validation portion. Combining the predictions from all three folds produces OOF probabilities for the complete training dataset.

Candidate thresholds are evaluated using the OOF probabilities, and the threshold that produces the highest F1-score is selected. The resulting threshold is 0.8752, with the unrounded value being 0.8751839224. The test set is not involved in this threshold-selection process.

The selected threshold is then fixed across all centralized and federated experiments. This ensures that differences in Precision, Recall, and F1-score are evaluated under the same decision rule rather than being caused by separately selecting thresholds for individual models.

The conventional 0.50 threshold is also evaluated later as a reference point, while 0.8752 remains the primary threshold for the experimental comparison.

Model	Threshold	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
Centralized (LR)	0.8752	95.51%	78.70%	67.33%	72.58%	0.9600	0.8170

Table 5.1 — Centralized Logistic Regression Performance

Table 5.1 shows the centralized baseline performance. The centralized model achieves 95.51% accuracy, 78.70% precision, 67.33% recall, and 72.58% F1-score, with ROC-AUC of 0.9600 and PR-AUC of 0.8170. These values establish the centralized Logistic Regression model as the reference baseline for evaluating the corresponding federated configurations.

5.1.3 IMPLEMENTATION OF FEDERATED LEARNING

The federated implementation uses Logistic Regression with FedProx-based local optimization. Unlike the centralized implementation, which uses `sklearn.LogisticRegression`, the federated implementation performs the Logistic Regression optimization directly so that the FedProx proximal term can be included in the local gradient update.

The model calculates the linear output and predicted probability as:

$$z = Xw + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

During local training, the weight gradient incorporates the Logistic Regression gradient, L2 regularization, and the FedProx proximal term:

$$\nabla_w = \frac{1}{m} X^T (\hat{y} - y) + \lambda w + \mu (w - w_{global})$$

The FedProx term penalizes excessive deviation of the local parameters from the current global parameters.

The server initializes the global parameters and sends them to the clients at the beginning of each communication round. Each client starts from the received global parameters and performs local training for 3 epochs using mini-batches of 128 samples. SMOTENC is applied locally before training, and synthetic samples remain within the client. After local training, the updated parameters are returned to the server.

The server combines the client parameters using sample-size-weighted averaging to produce the next global model. Thus, FedProx is responsible for local optimization, while sample-size-weighted averaging is responsible for server aggregation.

The federated implementation uses:

Parameter	Value
Local epochs	3
Batch size	128
Learning rate	0.10
L2 regularization	1×10^{-4}
FedProx μ	0.05
Communication rounds	40

Table 5.2 — Federated Learning Settings

Non-IID Client Distribution

For the primary non-IID experiment, the training data is distributed among three clients using class-wise Dirichlet partitioning with $\alpha = 0.30$. Diabetic and non-diabetic samples are partitioned separately. A smaller α produces greater skew, while a larger α produces more even distributions.

Client	Samples	Diabetic	Diabetic proportion
Client 1	62,357	2,495	4.00%
Client 2	4,765	4,282	89.86%
Client 3	9,794	9	0.09%

Table 5.3 — Non-IID Client Distribution

The Table 5.3 demonstrates strong client-level statistical heterogeneity. One client contains only 0.09% diabetic samples, while another contains 89.86%. This confirms that the non-IID experiment produces substantially different class distributions across clients.

IID Client Distribution

For the IID experiment, the training data is distributed using stratified allocation so that the clients maintain approximately the same class distribution as the overall training dataset.

Client	Samples	Diabetic	Diabetic Proportion
Client 1	25,639	2,262	8.82%
Client 2	25,639	2,262	8.82%
Client 3	25,638	2,262	8.82%

Table 5.4 — IID Client-Level Class Distribution

Table 5.4 shows the three IID clients have almost identical diabetic-class proportions of 8.82%. This provides a comparatively homogeneous client configuration and establishes the reference federated setting for comparison with the heterogeneous non-IID configuration.

Configuration	Accuracy	Precision	Recall	F1-score	ROC-AUC	PR-AUC
IID	95.49%	78.55%	67.16%	72.41%	0.9599	0.8168
Non-IID	95.65%	80.31%	67.10%	73.11%	0.9589	0.8150

Table 5.5 — Federated Learning Performance for Three Clients

Table 5.5 shows three-client federated model performance for both IID and non-IID data. The three-client federated results remain close to the centralized baseline despite the substantially different client distributions in the non-IID experiment. This demonstrates that, for the tested configuration, federated training can produce predictive performance comparable to centralized training without transferring the clients' raw training records to the server. However, this result represents the particular simulated configuration and should not be generalized to all federated settings.

5.1.4 IMPLEMENTATION OF THRESHOLD COMPARISON

The common threshold of 0.8752 is used as the primary decision rule for comparing the models. The conventional 0.50 threshold is additionally evaluated to demonstrate how the operating threshold affects Precision and Recall. The results are shown in Table 5.6.

Model	Clients	Accuracy	Precision	Recall	F1-score	ROC-AUC	PR-AUC
Centralized	—	88.43%	42.47%	87.85%	57.26%	0.9600	0.8170
IID	3	88.32%	42.20%	87.74%	56.99%	0.9599	0.8168
IID	5	88.38%	42.34%	87.62%	57.09%	0.9598	0.8170
IID	10	88.52%	42.62%	87.21%	57.26%	0.9594	0.8158
Non-IID	3	88.71%	43.02%	86.56%	57.48%	0.9589	0.8150
Non-IID	5	89.50%	44.96%	85.14%	58.84%	0.9563	0.8017
Non-IID	10	94.76%	70.04%	70.99%	70.51%	0.9567	0.8074

Table 5.6 — Results at the Conventional 0.50 Threshold

The conventional threshold produces considerably higher recall for most configurations but also produces lower precision. For example, the centralized model achieves 87.85% recall but only 42.47% precision at 0.50. The same general precision-recall trade-off is observed across the federated configurations. The comparison demonstrates why the project retains 0.8752 as the primary threshold rather than relying only on the conventional 0.50 threshold. The 0.50 threshold shifts the models toward higher recall and lower precision, whereas the fixed OOF-selected threshold provides the operating point used for the main centralized-versus-federated comparison.

5.1.5 IMPLEMENTATION OF CLIENT-COUNT EXPERIMENT

The client-count experiment investigates federated configurations containing 3, 5, and 10 clients under both IID and non-IID distributions. The training configuration, communication rounds, and evaluation procedure remain fixed while the number of clients is varied.

For IID experiments, stratified sampling is used. For non-IID experiments, class-wise Dirichlet partitioning with $\alpha = 0.30$ is used.

Distribution	Clients	Accuracy	Precision	Recall	F1-score	ROC-AUC	PR-AUC
IID	3	95.49%	78.55%	67.16%	72.41%	0.9599	0.8168
IID	5	95.50%	78.70%	67.10%	72.44%	0.9598	0.8170
IID	10	95.51%	78.65%	67.33%	72.55%	0.9594	0.8158
Non-IID	3	95.65%	80.31%	67.10%	73.11%	0.9589	0.8150
Non-IID	5	95.14%	74.74%	67.87%	71.14%	0.9563	0.8017
Non-IID	10	95.35%	97.51%	48.53%	64.80%	0.9567	0.8074

Table 5.7 — Federated Performance with Different Client Counts

Under IID partitioning, performance remains relatively stable as the number of clients increases. Accuracy changes only from 95.49% to 95.51%, while the other metrics also show relatively small changes.

The non-IID configurations show greater variation. At three clients, the model achieves 80.31% precision and 67.10% recall. At five clients, precision decreases to 74.74%, while at ten clients precision increases substantially to 97.51% but recall falls to 48.53%. The corresponding F1-score at ten clients is 64.80%.

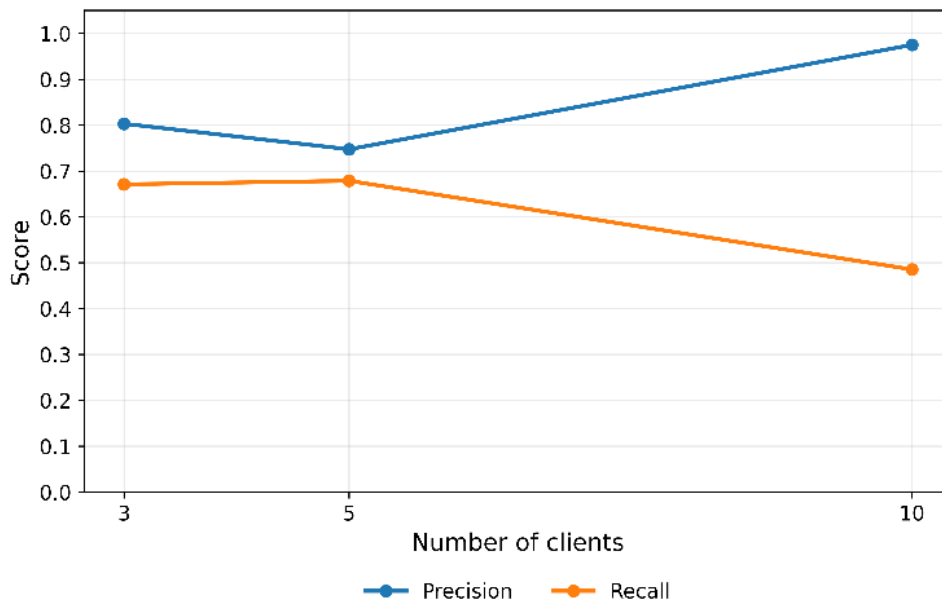


Fig 5.1. Precision-Recall Trend Across Non-IID Client Counts

Non-IID Clients	TP	FP	TN	FN
3	1,138	279	17,255	558
5	1,151	389	17,145	545
10	823	21	17,513	873

Table 5.8 — Non-IID Classification Counts

The classification counts explain the change observed at ten clients. The ten-client configuration produces only 21 false positives, compared with 279 for three clients and 389 for five clients. This contributes to its very high precision of 97.51%. However, it also produces 873 false negatives, considerably higher than the 558 and 545 false negatives for three and five clients, respectively. This explains the reduction in recall to 48.53%.

The client-count experiment shows that the IID configurations remain relatively stable across 3, 5, and 10 clients, whereas the non-IID configurations exhibit greater variation. However, the results should not be interpreted as showing that client count alone causes these changes. Changing the number of clients also changes the Dirichlet partition, client composition, and client sample sizes. Therefore, the experiment describes the behaviour of the particular generated configurations rather than isolating client count as an independent causal factor.

5.1.6 IMPLEMENTATION OF DIABETES PREDICTION APPLICATION

The project also includes a Streamlit-based diabetes prediction application that provides an interface for using the trained models.

The application allows the user to upload a medical report. The required patient attributes are extracted from the uploaded report and passed through the same feature-processing pipeline used by the trained model. The processed

values are then supplied to either the centralized or federated Logistic Regression model.

The application displays the resulting diabetes prediction to the user. In addition, SHAP-based explanations are provided to show the contribution of the input features toward the prediction. This provides an interpretable view of the model output rather than presenting only the final classification.

The application can demonstrate predictions using both the centralized model and the federated global model. This allows the project to show the practical use of the federated model after training, while the federated training process itself does not require raw client training records to be transferred to the server.

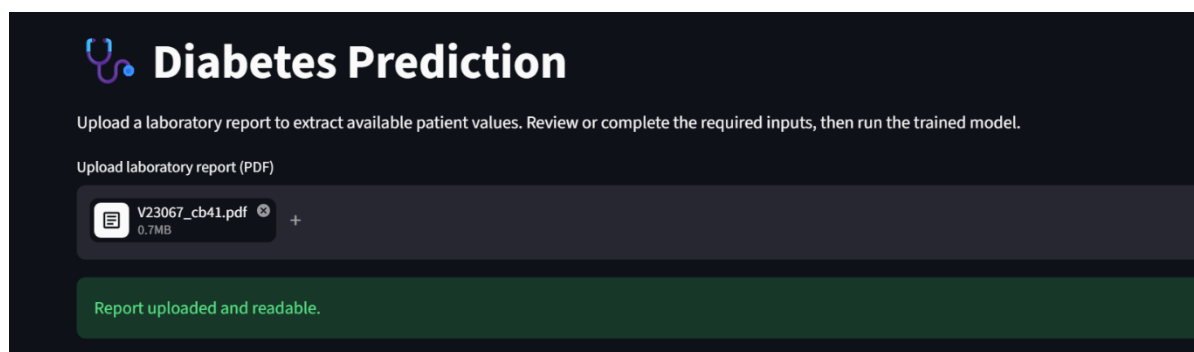


Fig 5.2. Medical Report Upload

The screenshot shows a form for entering patient attributes. It is organized into two columns. The left column contains: "Age" (77), "BMI" (30.00), "HbA1c (%)" (9.20), and "Blood glucose level (mg/dL)" (200.00). The right column contains: "Gender" (Female), "Hypertension" (No), "Heart disease" (No), and "Smoking history" (never). Each input field has a reset icon and minus/plus controls. Below the form, a note states: "The model uses age, BMI, HbA1c, blood glucose, hypertension, heart disease, gender, and smoking history." At the bottom is a large red button labeled "Predict Diabetes".

Fig 5.3. Extracted Patient Attributes



Fig 5.4. Model Prediction

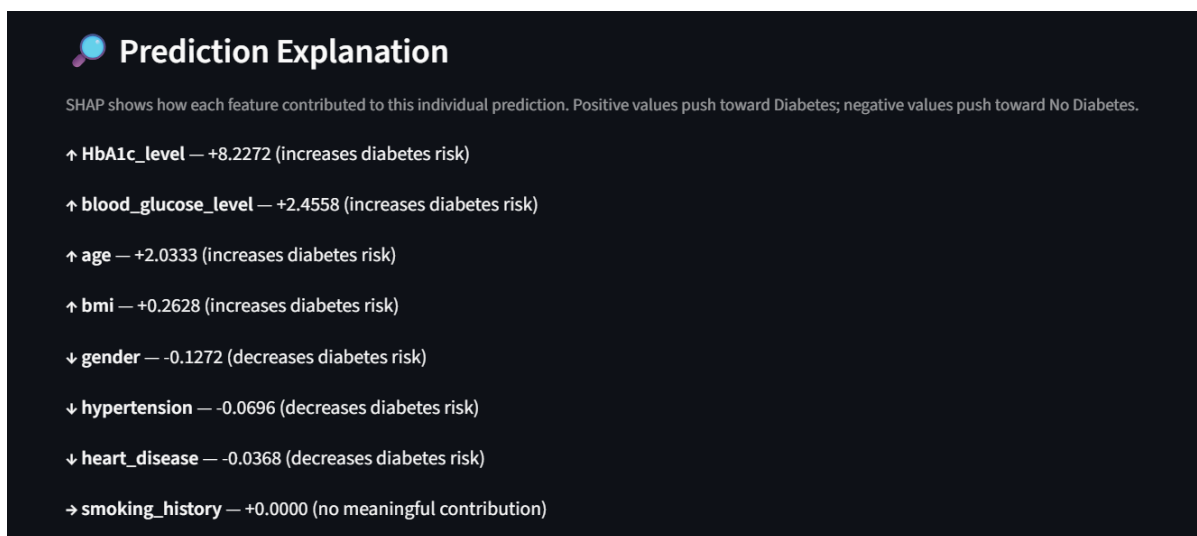


Fig 5.5. SHAP Explanation

5.2 DISCUSSION

The experimental results demonstrate that Logistic Regression can be used for diabetes prediction in both centralized and federated learning settings. The centralized model achieved an accuracy of 95.51%, while the three-client federated models achieved 95.49% accuracy under IID partitioning and 95.65% under non-IID partitioning at the common threshold of 0.8752. The corresponding F1-scores were also close, with 72.58% for the centralized model, 72.41% for the IID federated model, and 73.11% for the non-IID federated model. These results show that, for the three-client configurations evaluated in

this study, federated training produced predictive performance comparable to the centralized baseline.

The client distribution analysis demonstrates a clear difference between IID and non-IID settings. In the three-client IID configuration, the diabetic-class proportion was approximately 8.82% for each client. In contrast, the non-IID configuration produced diabetic-class proportions of 4.00%, 89.86%, and 0.09% across the three clients. Despite this substantial difference in client-level class distributions, the three-client non-IID model maintained performance close to the centralized and IID models. This indicates that the federated model was able to learn a useful global decision boundary even when the participating clients had substantially different class distributions.

The experiments with different numbers of clients show different behaviour under IID and non-IID distributions. IID performance remained relatively stable as the number of clients increased from 3 to 10, with accuracy remaining between 95.49% and 95.51% and F1-score remaining between 72.41% and 72.55%. In comparison, the non-IID configurations showed greater variation. The five-client configuration achieved 95.14% accuracy and 71.14% F1-score, while the ten-client configuration achieved 95.35% accuracy but had a precision of 97.51% and recall of only 48.53%.

The ten-client non-IID result needs to be interpreted carefully. The high precision is associated with only 21 false positives, while the model produced 873 false negatives. Therefore, although precision increased to 97.51%, recall decreased substantially because more diabetic cases were classified as non-diabetic. The result does not represent an improvement across all evaluation measures.

The classification threshold is also an important factor in interpreting these results. At the conventional threshold of 0.50, the centralized model achieved 87.85% recall but only 42.47% precision. At the common threshold of

0.8752, the centralized model achieved 78.70% precision and 67.33% recall. Similarly, for the ten-client non-IID model, the precision and recall changed substantially between the two thresholds. Therefore, part of the precision-recall variation observed in the experiments is related to the selected operating threshold in addition to differences in client data distributions. T036_Paper

The client-count results should also not be interpreted as evidence that client count alone causes the observed performance changes. In the non-IID experiments, changing the number of clients also changes the Dirichlet partition, client composition, and client sample sizes. Therefore, the experiments describe the behaviour of the particular generated federated configurations rather than isolating the independent causal effect of client count.

Overall, the experiments support the use of federated learning for the studied diabetes prediction task as an alternative training approach to centralized learning. The federated models achieved performance comparable to the centralized baseline in the evaluated configurations while the clients' raw training data was not transferred to the central server. However, the clients in this study were simulated by partitioning a single dataset, rather than representing independent real-world healthcare institutions. Therefore, the results need to be further validated using data from actual multi-institutional settings.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This project compared centralized and federated Logistic Regression for diabetes prediction, with a focus on the effect of client-level statistical heterogeneity on federated learning performance. The study evaluated federated learning under both IID and non-IID client distributions and investigated different numbers of participating clients. T036_Paper

The centralized model achieved an accuracy of 95.51%, while the three-client federated models achieved 95.49% accuracy under IID distribution and 95.65% under non-IID distribution using the common threshold of 0.8752. These results show that the federated models achieved performance comparable to the centralized baseline for the three-client configurations evaluated.

The experiments also showed that IID configurations remained relatively stable as the number of clients increased, whereas non-IID configurations exhibited greater variation in predictive performance. These variations were influenced by differences in client data distributions and the common classification threshold, and therefore cannot be attributed to client count alone.

Overall, the study demonstrates the feasibility of using federated learning for diabetes prediction while avoiding the transfer of clients' raw training data to a central server. However, since the clients were simulated from a single dataset, further validation using real multi-institutional healthcare data is required.

6.2 FUTURE WORK

The following areas can be considered for future development:

Real Multi-Institutional Data:

Future work can evaluate the proposed approach using data from multiple real healthcare institutions instead of simulated clients.

Differential Privacy:

Differential privacy can be incorporated to provide additional data privacy in the federated learning process.

Multi-Task Federated Learning:

Multi-task federated learning can be investigated for applications involving multi-disease diagnosis.

Federated Optimization for Heterogeneous Clients:

Further federated optimization techniques can be developed to better handle heterogeneous client settings.

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